Statistics Physics

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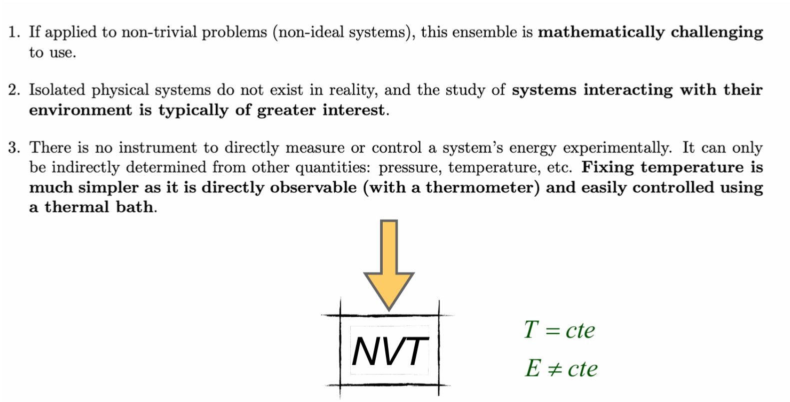
Contents

Chapter 1

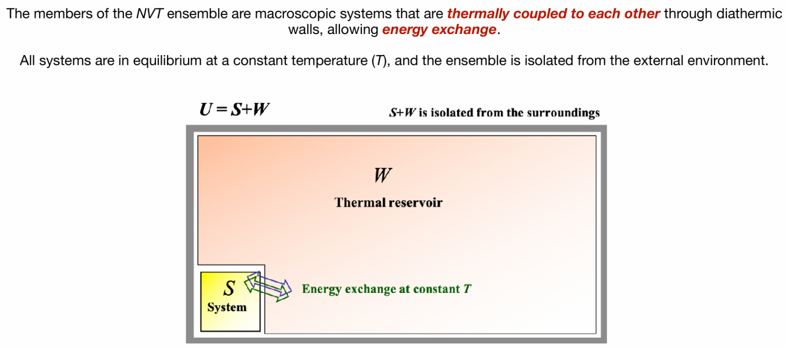
Introduction


1. If applied to non-trivial problems (non-ideal systems), this ensemble is **mathematically challenging** to use.
2. Isolated physical systems do not exist in reality, and the study of **systems interacting with their environment is typically of greater interest**.
3. There is no instrument to directly measure or control a system's energy experimentally. It can only be indirectly determined from other quantities: pressure, temperature, etc. **Fixing temperature is much simpler as it is directly observable (with a thermometer) and easily controlled using a thermal bath**.

In Chapter 4, we examined the microcanonical ensemble, focusing on isolated systems at equilibrium. In Chapter 5, we introduced the canonical ensemble, where systems in equilibrium exchange energy with their surroundings. This ensemble allows us to comprehend the behaviour of systems with a fixed energy content interacting with a thermal reservoir. Now, a key question arises: can we design an ensemble that captures the essence of macroscopic systems, ones that not only exchange energy with the external environment, but also mass? This scenario forms the basis of the macrocanonical (or grand canonical) ensemble, an ensemble tailor-made for systems achieving both energy and material equilibrium with the exterior. To this end, let's first resume the main conclusions on microcanonical and canonical ensembles.

- The **microcanonical ensemble** is mathematically challenging to employ. Moreover, the concept of isolated physical systems is more of an idealisation than a reality. In practical terms, the study of systems interacting with their environment often holds greater significance. Experimentally, there is no instrument available to directly measure or control a system's energy. Instead, it is typically inferred from other observables such as pressure, temperature, among others. By contrast, fixing the temperature proves to be more straightforward—it is directly observable through a thermometer and easily manageable with a thermal bath.

- The **canonical ensemble** proves to be a significantly more versatile ensemble, offering immense utility in addressing a multitude of physical problems within closed systems. However, the canonical ensemble exhibits significant limitations when attempting to address various other physical and chemical problems: *(1)* systems in equilibrium that exchange particles with the surroundings (open systems); *(2)* systems in equilibrium where chemical reactions take place; and *(3)* systems separated into phases in mutual coexistence (liquid-vapour, solid-vapour, etc.). These are all systems where the number of particles of each species is not constant.



Let's consider an isolated system at equilibrium. This system is formed by two subsystems, A and B, that can exchange energy and mass, as shown in Fig. 6.1. If the chemical potential of A, μ_A, equals the chemical potential of B, μ_B, then there is no net mass exchange. However, if this is not the case, then there will be a mass transfer from A to B if $\mu_A > \mu_B$ or from B to A if $\mu_A < \mu_B$. We should note that this mass transfer is internal to the A+B system. In other words,

$$N_A + N_B = \text{const} \rightarrow dN_A + dN_B = 0 \tag{6.1}$$

Additionally, the total entropy change of the system reads

$$dS = \left(\frac{\partial S}{\partial N_A}\right)_{N_B,U,V} dN_A + \left(\frac{\partial S}{\partial N_B}\right)_{N_A,U,V} dN_B = dN_A \left[\left(\frac{\partial S}{\partial N_A}\right) - \left(\frac{\partial S}{\partial N_B}\right)\right] \tag{6.2}$$

where the infinitesimal entropy change with mass is proportional to the chemical potential:

$$\mu = -T \left(\frac{\partial S}{\partial N}\right)_{U,V} \tag{6.3}$$

Therefore, substituting Eq. 6.3 into Eq. 6.2, we obtain

$$dS = -\frac{dN_A}{T}(\mu_A - \mu_B) \tag{6.4}$$

Note that, if $\Delta\mu \equiv \mu_A - \mu_B > 0$, then $dN_A < 0$; by contrast, if $\Delta\mu < 0$, then $dN_A > 0$. In both cases, $dS > 0$ and $dS = 0$ if no mass transfer occurs, that is if $\Delta\mu = 0$.

Figure 6.2: System where a chemical reaction takes place and a product forms.

Now, let's consider a system where the generic chemical reaction A + B = C occurs (see Fig. 6.2). At equilibrium, the chemical potentials of the three species are related as follows

$$\mu_A + \mu_B = \mu_C \tag{6.5}$$

Out of equilibrium, then products or reagents can form:

$$\mu_A + \mu_B > \mu_C \implies \text{A} + \text{B} \rightarrow \text{C} \tag{6.6}$$

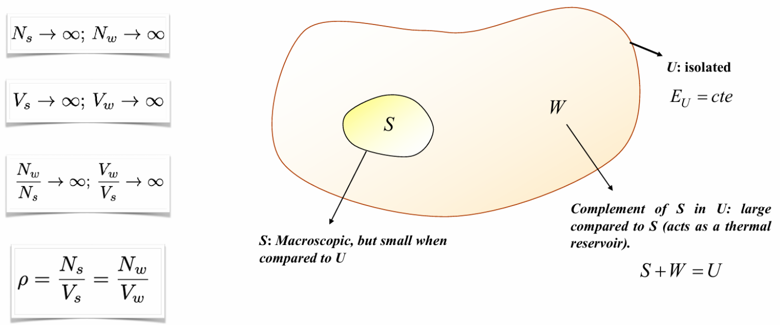
$$\mu_A + \mu_B < \mu_C \implies \text{C} \rightarrow \text{A} + \text{B} \tag{6.7}$$

In Fig. 6.3, we schematically show a representative member S of the macrocanonical ensemble that exchange energy and mass with its surrounding reservoir W. The net mass exchange is zero at the thermodynamic equilibrium, when $T_s = T_w$ and $\mu_s = \mu_w$. In particular, S is confined within rigid, permeable and diathermic walls that allow mass and energy transfer, but leave temperature (T), volume (V) and chemical potential (μ) constant. To stress these conditions, the macrocanonical ensemble that we wish to design is also denoted as (μ, V, T) ensemble and the corresponding partition function as $\Xi = \Xi(\mu, V, T)$. Additionally, we will introduce the **macrocanonical potential**, also known as the grand canonical potential or the Helmholtz free energy in the macrocanonical ensemble, $J(\mu, V, T) = -kT \ln \Xi$, defined as:

$$J = U - TS - \mu N = A - \mu N \tag{6.8}$$

where U is the internal energy, S the entropy, N the number of particles and A the Helmholtz free energy. The minimisation of the macrocanonical potential at equilibrium conditions provides the most probable distribution of energy and particles for a given temperature, chemical potential, and volume.

1.1 Design of the grand canonical ensemble

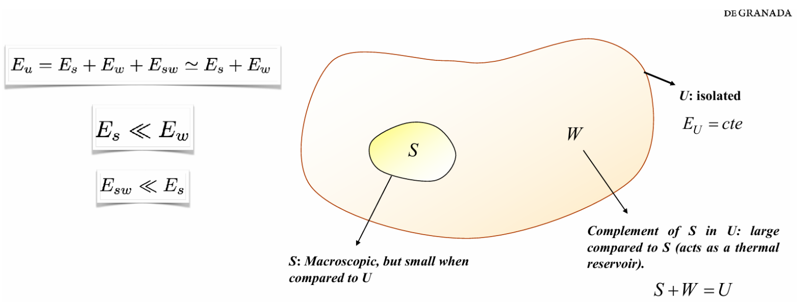


With reference to Fig. 6.4, the system S is very large, but much smaller than W, that is $S \ll W$, and

$$1 \ll N_s \ll N_w \tag{6.9}$$

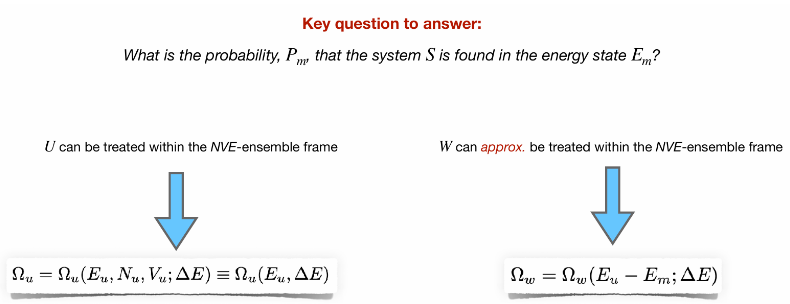
Additionally, the numerical density is basically constant across all U:

$$\frac{N_s}{V_s} \simeq \frac{N_w}{V_w} \tag{6.10}$$



If we disregard the interaction energy between S and W, the energy of the "supersystem" U reads

$$E_u \simeq E_s + E_w, \quad \text{with } E_s \ll E_w \tag{6.11}$$



Given that the Hamiltonian of the super-system U is separated into two parts $(S + W)$, the Schrödinger equation also separates:

$$\hat{H}_s|\phi_s\rangle = E_s|\phi_s\rangle \tag{6.12}$$

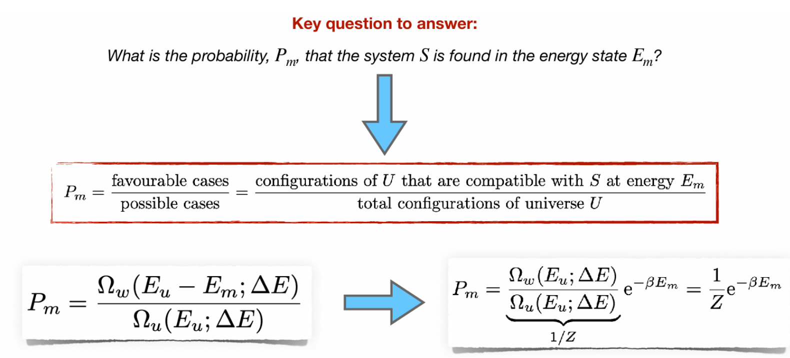
$$\hat{H}_w|\phi_w\rangle = E_w|\phi_w\rangle \tag{6.13}$$

where $E_s(N)$ is the energy of S, E_w is the energy of W and $|\phi_u\rangle = |\phi_s\rangle|\phi_w\rangle$. Given that S has quantum numbers that are independent on W, it makes sense to estimate the probability, P_{mN}, that S contains N particles and is in the energy state E_{mN}. Since W is much larger than S, we can also consider W as an isolated system with energy $E_w = E_u - E_{mN}$ and $N_w = N_u - N$ particles. It follows that the number of accessible states in U reads

$$\Omega_u = \Omega_u(E_u, N_u; \Delta E) \tag{6.14}$$

and the number of accessible states in U compatible with S containing N particles and having energy E_{mN}, reads

$$\Omega_w = \Omega_w(E_u - E_{mN}, N_u - N; \Delta E) \tag{6.15}$$



We can then calculate the aforementioned probability in terms of the energy and number of particles of the reservoir W:

$$P_{mN} = \frac{\text{favourable cases}}{\text{possible cases}} = \frac{\Omega_w(E_u - E_{mN}, N_u - N; \Delta E)}{\Omega_u(E_u, N_u; \Delta E)} \tag{6.16}$$

Given that $E_{mN} \ll E_u$, we can develop a power series of Ω_w. Nevertheless, since Ω_w increases exponentially with energy and converges slowly, we prefer to Taylor-expand the logarithm of Ω_w near the point (E_u, N_u):

$$\ln \Omega_w(E_w, N_w; \Delta E) \simeq \ln \Omega_w(E_u, N_u; \Delta E) + \left(\frac{\partial \ln \Omega_w}{\partial E_w}\right)_{E_w = E_u} (E_w - E_u) + \left(\frac{\partial \ln \Omega_w}{\partial N_w}\right)_{N_w = N_u} (N_w - N_u) \tag{6.17}$$

or equivalently

$$\ln \Omega_w(E_w, N_w; \Delta E) \simeq \ln \Omega_w(E_u, N_u; \Delta E) - \underbrace{\frac{\partial \ln \Omega_w}{\partial E_w}}_{\beta = 1/kT} E_{mN} - \underbrace{\frac{\partial \ln \Omega_w}{\partial N_w}}_{-\beta\mu} N \tag{6.18}$$

where we used the fact that the thermal bath W can be approximately treated as an isolated system and therefore within the microcanonical-ensemble framework. It follows that $S_w = k \ln \Omega_w$ and, from Eq. 6.3, we obtain the chemical potential in the above equation. Therefore,

$$\Omega_w(E_u - E_{mN}, N_u - N; \Delta E) \simeq \Omega_w(E_u, N_u; \Delta E) e^{-\beta(E_{mN} - \mu N)} \tag{6.19}$$

and the probability reads

$$P_{mN} = \underbrace{\frac{\Omega_w(E_u, N_u; \Delta E)}{\Omega_u(E_u, N_u; \Delta E)}}_{\leq 1} e^{-\beta(E_{mN} - \mu N)} \tag{6.20}$$

To calculate P_{mN}, we can impose the normalisation condition:

$$\sum_{N=0}^{\infty} \sum_m P_{mN} = 1 \tag{6.21}$$

which gives

$$\sum_{N=0}^{\infty}\sum_{m}\underbrace{\frac{\Omega_w(E_u,N_u;\Delta E)}{\Omega_u(E_u,N_u;\Delta E)}}_{\equiv 1/\Xi}e^{-\beta(E_{mN}-\mu N)}=1 \tag{6.22}$$

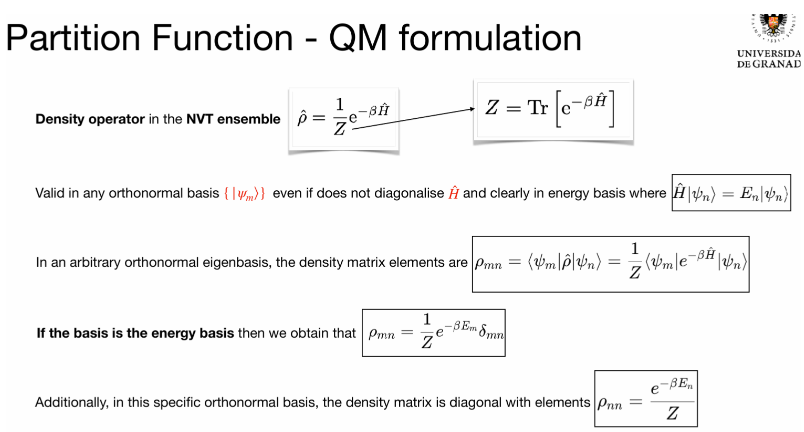
where the sum is over all possible states and number of particles of system S, without restriction. Rearranging, one obtains

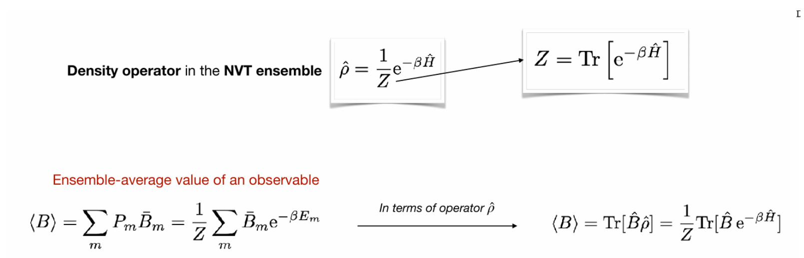
$$\Xi \equiv \frac{\Omega_u(E_u,N_u;\Delta E)}{\Omega_w(E_u,N_u;\Delta E)}=\sum_{N=0}^{\infty}\sum_{m}e^{-\beta(E_{mN}-\mu N)} \tag{6.23}$$

Substituting this result into Eq. 6.20, we obtain

$$P_{mN}=\frac{1}{\Xi}e^{-\beta(E_{mN}-\mu N)} \tag{6.24}$$

where $\Xi=\Xi(\mu,T,V)$ is the **macrocanonical partition function** and P_{mN} is the **macrocanonical probability**. Note that both Ξ and P_{mN} do not depend on W.





In terms of operators:

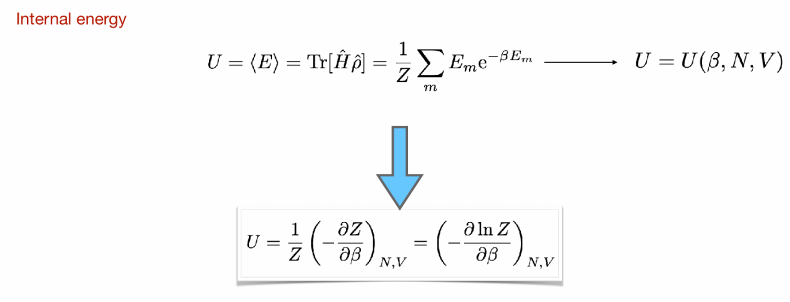
$$\hat{\rho} = \frac{1}{\Xi} e^{-\beta(\hat{H}-\mu\hat{N})} \tag{6.25}$$

$$\Xi = \mathrm{Tr}\left[e^{-\beta(\hat{H}-\mu\hat{N})}\right] \tag{6.26}$$

and the average value of a generic observable reads

$$\langle b \rangle = \mathrm{Tr}\left[\hat{b}\hat{\rho}\right] = \frac{1}{\Xi}\mathrm{Tr}\left[\hat{b}e^{-\beta(\hat{H}-\mu\hat{N})}\right] \tag{6.27}$$

1.2 Connection with Thermodynamics



In this section, our goal is understanding what the physical meaning of μ is and how Ξ, and more generally the macrocanonical ensemble, is related to Thermodynamics. In the (μ, V, T) ensemble, both E and N are not constant as the system exchanges energy and mass with the surroundings. Nevertheless, from a Thermodynamics perspective, we can still define their average values:

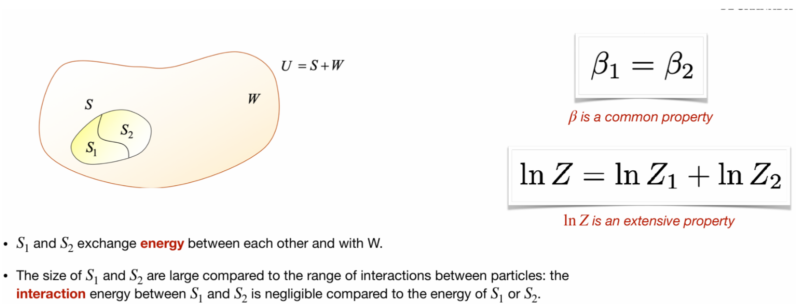
$$U = \langle E \rangle = \sum_{N=0}^{\infty}\sum_m E_{mN}P_{mN} = \frac{1}{\Xi}\sum_{N=0}^{\infty}\sum_m E_{mN}e^{-\beta(E_{mN}-\mu N)} \tag{6.28}$$

where E_{mN} is also a function of the volume V. Let's define the **fugacity** as $\lambda \equiv e^{\beta\mu}$. Then Eq. 6.28 is modified as

$$U = \frac{1}{\Xi}\sum_{N=0}^{\infty}\sum_m E_{mN}e^{-\beta E_{mN}}\lambda^N = -\frac{1}{\Xi}\sum_{N=0}^{\infty}\lambda^N\sum_m \left.\frac{\partial e^{-\beta E_{mN}}}{\partial \beta}\right|_V = -\frac{1}{\Xi}\frac{\partial}{\partial \beta}\left[\sum_{N=0}^{\infty}\lambda^N\sum_m e^{-\beta E_{mN}}\right]_{\lambda,V} \tag{6.29}$$

which can simply be written as

$$U = -\frac{1}{\Xi}\left(\frac{\partial \Xi}{\partial \beta}\right)_{\lambda,V} = -\left(\frac{\partial \ln \Xi}{\partial \beta}\right)_{\lambda,V} \tag{6.30}$$



It then follows that

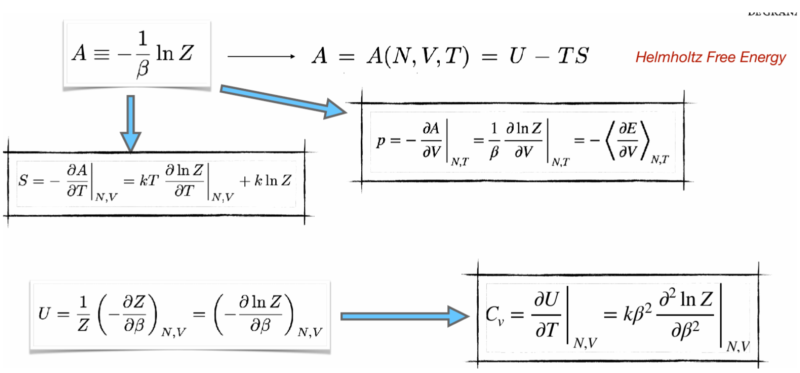
$$\beta_1 = \beta_2 = \beta, \quad \mu_1 = \mu_2 = \mu \tag{6.36}$$

This suggests that μ does not depend on the system size or, in other words, that it is an **intensive** property. It also follows that

$$\Xi = \Xi_1 \Xi_2 \tag{6.37}$$

or

$$\ln \Xi = \ln \Xi_1 + \ln \Xi_2. \tag{6.38}$$



This result indicates that the property $\ln \Xi$ and hence also $J \equiv -kT \ln \Xi$, which we call macrocanonical potential, are **extensive** variables. In particular, the macrocanonical potential depends on temperature, chemical potential, and energy state of the system. Mathematically:

$$J = J(\beta, \mu, \{E_{mN}\}) \tag{6.39}$$

Its total differential can therefore be written as

$$d(\beta J) = \left.\frac{\partial(\beta J)}{\partial \beta}\right|_{\mu,\{E_{mN}\}} d\beta + \left.\frac{\partial(\beta J)}{\partial \mu}\right|_{\beta,\{E_{mN}\}} d\mu + \sum_{N=0}^{\infty}\sum_{m} \left.\frac{\partial(\beta J)}{\partial E_{mN}}\right|_{\beta,\mu} dE_{mN} \quad (6.40)$$

Let's calculate these three partial derivatives separately.

I

$$\begin{aligned}
\left.\frac{\partial(\beta J)}{\partial \beta}\right|_{\mu,\{E_{mN}\}} &= -\frac{\partial \ln \Xi}{\partial \beta} = -\frac{1}{\Xi}\left.\frac{\partial \Xi}{\partial \beta}\right|_{\mu,\{E_{mN}\}} = \\
&= -\frac{1}{\Xi}\sum_{N=0}^{\infty}\sum_{m}(-E_{mN}+\mu N)e^{-\beta(E_{mN}-\mu N)} = \\
&= \frac{1}{\Xi}\sum_{N=0}^{\infty}\sum_{m} E_{mN}e^{-\beta E_{mN}+\beta\mu N} - \mu\frac{1}{\Xi}\sum_{N=0}^{\infty}\sum_{m} N e^{-\beta E_{mN}+\beta\mu N} = \\
&= \langle E\rangle - \mu\langle N\rangle = \\
&= U - \mu\langle N\rangle \qquad (6.41)
\end{aligned}$$

II

$$\left.\frac{\partial(\beta J)}{\partial \mu}\right|_{\beta,\{E_{mN}\}} = -\left.\frac{\partial \ln \Xi}{\partial \mu}\right|_{T,V} = -\beta\langle N\rangle \qquad (6.42)$$

III

$$\left.\frac{\partial(\beta J)}{\partial E_{mN}}\right|_{\beta,\mu} = -\frac{1}{\Xi}\frac{\partial \Xi}{\partial E_{mN}} = -\frac{1}{\Xi}(-\beta)e^{-\beta E_{mN}+\beta\mu N} = \beta P_{mN} \qquad (6.43)$$

Therefore Eq. 6.40 is re-written as

$$d(\beta J) = (U - \mu\langle N\rangle)d\beta - \beta\langle N\rangle d\mu + \beta\sum_{Nm} P_{mN} dE_{mN} \qquad (6.44)$$

where $\sum_{Nm} \equiv \sum_{N=0}^{\infty}\sum_{m}$. We now note that:

$$\begin{aligned}
U d\beta &= d(U\beta) - \beta dU \\
-\mu\langle N\rangle d\beta - \beta\langle N\rangle d\mu &= -d(\beta\mu\langle N\rangle) + \beta\mu d\langle N\rangle \qquad (6.45)
\end{aligned}$$

It follows that

$$d(\beta J) = d(U\beta) - \beta dU - d(\beta\mu\langle N\rangle) + \beta\mu d\langle N\rangle + \beta\sum_{Nm} P_{mN} dE_{mN} \tag{6.46}$$

and rearranging, we obtain

$$d[\beta(U - \mu\langle N\rangle - J)] = \beta dU - \beta\mu d\langle N\rangle - \beta\sum_{Nm} P_{Nm} dE_{mN} \tag{6.47}$$

Now, what is the physical meaning of the last term in the r.h.s. of the above equation? To answer this question, we first notice that we are assuming that $E_{mN} = E_{mN}(V)$ and, if the volume changes, then the energy will change accordingly. In other words, dE_{mN} is a change in the energy levels due to a volume change dV. If we impose a volume change, keeping constant all the other variables, the internal energy will also change:

$$U = \sum_{Nm} P_{Nm} E_{Nm} \rightarrow U' = \sum_{Nm} P_{Nm} E'_{Nm} \tag{6.48}$$

If we now apply the First Law of Thermodynamics, we observe that

$$dU = U' - U = \sum_{Nm} P_{Nm} dE_{Nm} = \sum_{Nm} P_{Nm}\left(\frac{\partial E_{Nm}}{\partial V}\right) dV = W = -pdV \tag{6.49}$$

where p stands for pressure. It becomes clear that

$$-\sum_{Nm} P_{Nm}\left(\frac{\partial E_{Nm}}{\partial V}\right) = p, \quad \text{pressure} \tag{6.50}$$

and

$$\sum_{Nm} P_{Nm} dE_{Nm} = W, \quad \text{mechanical work.} \tag{6.51}$$

If we incorporate this result into Eq. 6.52, then we obtain

$$d[\beta(U - \mu\langle N\rangle - J)] = \beta[dU - \mu d\langle N\rangle - W] = \beta[dU + pdV - \mu d\langle N\rangle] \tag{6.52}$$

Equivalently

$$d\left[\frac{1}{T}(U - \mu\langle N\rangle - J)\right] = \frac{1}{T}[dU + pdV - \mu d\langle N\rangle] \tag{6.53}$$

Let's now apply the Second Law of Thermodynamics and obtain the Fundamental Equation of Thermodynamics, which combines the first and the second laws:

$$Q = TdS = dU + pdV - \mu dN \tag{6.54}$$

Rearranging, we obtain the infinitesimal change of entropy:

$$dS = \frac{1}{T}[dU + pdV - \mu dN] \tag{6.55}$$

or the infinitesimal change of internal energy:

$$dU = TdS - pdV + \mu dN \tag{6.56}$$

Comparing Eq. 6.55 with Eq. 6.53, one observes that μ is indeed the chemical potential and that the argument of the derivative in the l.h.s. of Eq. 6.53 results to be

$$U - \mu\langle N\rangle - J = TS \tag{6.57}$$

or, rearranging, that

$$J = U - TS - \mu\langle N\rangle \tag{6.58}$$

which justifies the definition of macrocanonical potential for $J(\mu, T, V) = -kT\ln\Xi$. Additionally, from Eq. 6.56, we see that

$$U = TS - pV + \mu N \tag{6.59}$$

and finally

$$J(\mu, T, V) = U - TS - \mu N = -pV = -p(T, \mu)V \tag{6.60}$$

This last equation, combined with $J = -kT \ln \Xi$, is crucial to derive all the equations of Thermodynamics. More specifically:

$$S = -\left.\frac{\partial J}{\partial T}\right|_{\mu,V} = kT \left.\frac{\partial \ln \Xi}{\partial T}\right|_{\mu,V} + k \ln \Xi, \quad \text{Entropy} \tag{6.61}$$

$$p = -\left.\frac{\partial J}{\partial V}\right|_{\mu,T} = kT \left.\frac{\partial \ln \Xi}{\partial V}\right|_{\mu,T} = \frac{kT}{V} \ln \Xi, \quad \text{Pressure} \tag{6.62}$$

$$\langle N \rangle = -\left.\frac{\partial J}{\partial \mu}\right|_{T,V} = kT \left.\frac{\partial \ln \Xi}{\partial \mu}\right|_{T,V}, \quad \text{Average number of particles} \tag{6.63}$$